



GROUP- 5

A Project Proposal on CapTrack: Capstone Project Lifecycle Management System

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CapTrack: Capstone Project Lifecycle Management System

Introduction

Final Year Undergraduate Capstone Projects play a vital role in a student's academic journey. However, the overall process of forming project groups, selecting supervisors, managing project progress, and maintaining documentation is often fragmented across multiple platforms such as social media, email, messaging applications, and cloud storage services.

Students frequently face difficulties finding suitable teammates with complementary skills, while supervisors struggle to identify qualified students who match their research interests. In addition, academic departments often lack a centralized system for monitoring supervisor workloads, available supervision slots, project histories, and student progress.

To address these challenges, **CapTrack** is proposed as a comprehensive web-based platform that manages the complete capstone project lifecycle, from student profiling and team formation to project execution, progress monitoring, evaluation, and documentation. The system provides a centralized environment where students, supervisors, and administrators can efficiently collaborate throughout the entire capstone process.

Problem Statement

The existing capstone project management process suffers from several limitations:

- Lack of a centralized platform for student group formation.
- Difficulty in matching students with appropriate supervisors.
- Limited visibility of faculty research interests and supervision availability.
- Difficulty in tracking supervisors' research domains and accessing records of their previous capstone groups and students.
- Project communication scattered across multiple external platforms.
- Inefficient tracking of project tasks, milestones, submissions, and feedback.
- Manual administrative monitoring of supervisor workloads and student progress.

These issues reduce efficiency, create communication gaps, and limit transparency throughout the capstone project lifecycle.

Project Objectives

The primary objectives of CapTrack are:

1. To provide a centralized platform for capstone project management.
2. To facilitate efficient student-to-student and student-to-supervisor matching.
3. To maintain a comprehensive repository of faculty information and past project records.
4. To improve communication among students, supervisors, and administrators.
5. To support project monitoring through task assignment and progress tracking.
6. To enable supervisors to efficiently evaluate projects, provide feedback, and track the real-time progress and individual contributions of group members.

System Modules and Features

CapTrack will function as an integrated Capstone Project Lifecycle Management System that combines student profiling, collaboration, project management, and administrative monitoring into a single web application.

The system will support three primary user roles:

- Student
- Supervisor
- Administrator

Each role will have dedicated functionalities designed to streamline capstone project activities.

1. Student Profile Management

This module assists students in finding suitable teammates while providing transparency regarding teammates details information. Here, students will create detailed profiles containing:

i. Academic Information

Each profile will include academic details such as the student's identity, department, academic standing, target capstone semester, completed credits, expected credits before beginning the capstone project, and overall academic performance.

ii. Technical and Professional Skills

In addition to academic information, students will be able to showcase their technical and professional competencies, including programming knowledge, familiarity with technologies and frameworks, research capabilities, documentation experience, communication skills, and leadership qualities.

iii. External Profiles

The profile will also provide space for linking external professional platforms such as GitHub repositories, LinkedIn accounts, and personal portfolio websites so that supervisors and potential teammates can review previous work and achievements.

iv. Research Interests

Students can provide a detailed statement describing their interests, preferred project domains, and future career goals.

Advanced Search and Filtering

The platform will include a search and filtering mechanism that allows users to discover suitable teammates and supervisors based on various academic and professional criteria. Students and supervisors will be able to explore profiles by considering factors such as academic performance, department affiliation, technical skills, research interests, target capstone semester, and academic progress. By providing flexible search capabilities, this feature will simplify the process of identifying compatible collaborators and selecting supervisors whose expertise aligns with specific project goals and research interests.

2. Supervisor Profiles and History

This module will maintain a comprehensive supervisor database including:

i. Supervisor Details

Each profile contains essential professional details, including the supervisor's name, email address, designation, explicit research interests, and core areas of expertise.

ii. Supervision Status

To ensure transparency during the group formulation phase, the system displays a live supervision status for each faculty member. Supervisors can easily update their current intake capacity, allowing students to instantly view available slots, remaining capacity, or whether a supervisor's quota is completely full for the upcoming semester(s).

iii. Previous Project Insights

The system incorporates a robust historical archive that meticulously stores project information from previous semesters, including project titles, detailed abstracts, team member compositions, assigned supervisors, and specific semester timelines. By keeping a comprehensive record of these past works, the archive serves as a transparent reference point that helps reduce project

duplication and encourages high-value academic innovation across the department. Furthermore, this historical data provides incoming students with a clear, practical understanding of each supervisor's ongoing research trajectory and the specific types of projects they typically mentor, making it much easier to choose a supervisor whose practical expertise aligns with their goals.

3. Collaboration Feed

The Collaboration Feed acts as a centralized networking and communication hub within the platform, bridging the gap between students and supervisors to create a transparent, dynamic, and efficient project matchmaking environment.

i. Supervisor Recruitment Posts

Supervisors seeking specific student groups can actively publish project opportunities on the feed. When creating a post, they can define clear parameters by specifying the intended project topic, research area, expected team size, and required technical skills, alongside any additional prerequisites. This allows faculty members to pitch unique research concepts directly to the student body and recruit talent that matches their exact project needs.

ii. Student Recruitment Posts

Similarly, students who have already formed partial teams but are missing critical skill sets can post targeted recruitment requests. In these posts, students can outline their current team size, the overarching project idea, preferred technologies, and the specific skills they are looking for in a new member. This functionality helps incomplete groups find the right peers without relying on random matching or informal social networks.

4. Capstone Progress and Task Management Module

In spite of project group matching, the platform will provide a dedicated workspace for each project team. This centralized workspace ensures that all project-related activities are tracked in one secure environment.

i. Group Workspace

Each project group is provisioned with a dedicated dashboard that serves as the central hub for the team. This workspace displays comprehensive team information, grants direct access to the assigned supervisor, and provides a real-time progress overview. This layout ensures that both students and faculty have immediate visibility into the overall health and status of the project.

ii. Task Assignment and Milestone Management

Supervisors can seamlessly create and manage essential academic milestones, such as proposal submissions, literature reviews, SRS documentation, sprint reviews, prototype demonstrations, and final report submissions. Tasks can be flexibly assigned to individual students to track personal accountability or to the entire group for collaborative work. Each task entry allows the supervisor to specify a detailed description, a hard deadline, priority levels, and explicit submission requirements.

iii. Submission Management

To fulfill assigned tasks, students can upload or link their deliverables directly through the platform. The system supports a wide range of formats, allowing students to submit documents, source code archives, presentation slides, external repository links, or cloud storage links. Furthermore, the system automatically maintains a complete submission history and version records, preventing data loss and ensuring that the evolution of the project is documented.

iv. Feedback and Evaluation

Supervisors are equipped with built-in tools to review submissions, provide detailed written feedback, request necessary revisions, and approve deliverables once they meet the required standards. Additionally, faculty can assign scores and grades directly within the interface. This module eliminates the need for external tools by creating a highly structured, transparent, and auditable evaluation process entirely within the platform.

Expected Benefits

The implementation of this platform provides a unified digital ecosystem that significantly improves efficiency, accountability, and quality throughout the capstone project lifecycle. The key advantages of adopting this system are structured below:

i. Enhanced Transparency

Administrators gain real-time visibility into the entire capstone process without relying on manual tracking or spreadsheets. From a central dashboard, they can monitor student participation, balance supervisor workloads, check group formation status, and track live project progress.

ii. Improved Project Quality & Innovation

By maintaining a searchable historical archive, the system provides a clear reference point for the department. Students can easily review past work to avoid duplicating old project ideas, encouraging them to pursue unique and innovative research topics.

iii. Optimized Team Formation

The system eliminates random pairing or reliance on informal social circles. Students can systematically search for and connect with suitable teammates based on compatible skills, academic backgrounds, and shared research interests, resulting in stronger, well-balanced teams.

iv. Comprehensive Institutional Repository

Every task submission, revision history, supervisor review, and individual contribution is permanently archived. This creates a secure academic data repository and a valuable institutional knowledge base to support future students, faculty, and academic coordinators.

Contribution to Academia

CapTrack contributes significantly to academic project management by:

1. Providing data-driven transparency for project allocation and supervision.
2. Supporting informed decision-making through historical project records.
3. Promoting equitable opportunities for students regardless of networking ability.
4. Reducing administrative workload through automation.
5. Improving collaboration among students and faculty.
6. Establishing a complete digital repository of capstone project activities and outcomes.
7. Preserving project knowledge and student contributions for future academic use

Conclusion

CapTrack is a comprehensive Capstone Project Lifecycle Management System designed to modernize and streamline the management of undergraduate capstone projects and thesis activities. By integrating student profiling, supervisor matching, collaboration, project tracking, evaluation, and documentation into a single platform, the system addresses many of the challenges currently faced by students, supervisors, and academic administrators.

The proposed solution will enhance transparency, improve project quality, facilitate collaboration, and create a sustainable digital ecosystem for managing capstone projects throughout their entire lifecycle.